

思考 10.12

⑩ 三个不动点都确定一个映射了,

唯一性。恒等映射, 过三不动点

∴ 只能是恒等映射

⑫ ~~设~~ $w(z) = z^n$ 幂函数映射

带 $\rightarrow$ 角 指数

角 $\rightarrow$ ~~带~~ 对数

5(1) 6(2) 7(1) 10

$$5(1) W=f(z)=e^{i\theta} \frac{z-i}{z+i}$$

$$W'=e^{i\theta} \frac{z-i}{(z+i)^2} \quad W(i)=e^{i\theta}(-\frac{i}{2})$$

$$\arg W(i) = \theta + \frac{\pi}{2} = -\frac{\pi}{2} \Rightarrow \theta = -\pi$$

$$f(z) = -\frac{z-i}{z+i}$$

$$6(2). w=f(z)=e^{i\varphi} \frac{z-z_0}{1-z\bar{z}_0}$$

$$z_0 = \frac{1}{2}$$

$$w(-1) = e^{i\varphi} \frac{-1-\frac{1}{2}}{1-(-1)\frac{1}{2}} = -e^{i\varphi} = 1$$

$$\varphi = -\pi$$

$$\therefore W = \frac{z-\frac{1}{2}}{1-\frac{1}{2}z}$$

$$7(1) \frac{z-i}{z-(-2)} \frac{z-(-2)}{z-i} = \frac{1-i}{1-(-1)} \cdot \frac{W-(-1)}{W-i}$$

$$\Rightarrow \frac{z-i}{4} \frac{z+2}{z-i} = \frac{1-i}{2} \frac{W+1}{W-i}$$

